MTE 504 LECTURE NOTE 4

4.0 SILICON BULK AND SURFACE MICROMACHINING TECHNOLOGY FOR MICRO SYSTEMS OR MEMS

MEMS micromachining techniques allow for the construction of three-dimensional (3D) microsized structures, components, and various elements on or within a substrate (usually silicon). In some cases, micromachining is the utilization of modified IC manufacturing processes in conjunction with other processes such as deep bulk etching, laser assisted chemical vapour deposition, electroplating, and molding techniques.

Three widely used MEMS fabrication methods are

- surface micromachining,
- bulk micromachining, and
- LIGA (Lithography, Galvanoformung (electroforming), and Abformung (molding).

Below are scanning electron microscope (SEM) images of products from each type of micromachining process. The far left SEM shows microchambers and channels fabricated using bulk micromachining. The middle SEM shows layers of gears made possible through surface micromachining. The left SEM is a waveguide produced by Sandia National Laboratories using LIGA.

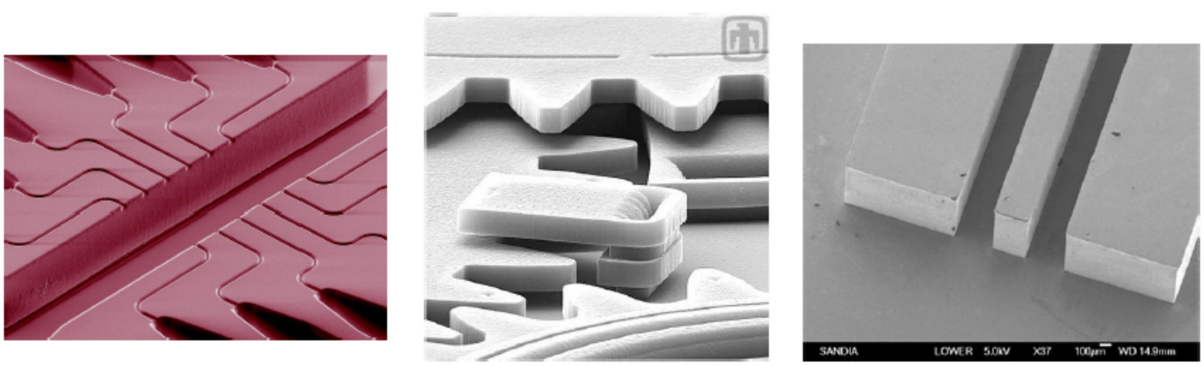


Figure 4.0: Products from micromachining processes

4.1 Overview the application of Surface and Bulk Micromachining and LIGA

Surface micromachining constructs thin mechanical components and systems on the surface of a substrate by alternately depositing, patterning and etching thin films. Bulk micromachining etches into a substrate to form 3D mechanical elements such as channels, chambers and valves. When combined with wafer bonding, surface and bulk micromachining allow for the fabrication of complex mechanical devices. LIGA processes combine collimated x-ray lithography with electroplating and molding techniques to create high aspect ratio (tall and thin) structures or deep cavities needed for certain types of MEMS devices.

4.2 Surface Micromachining: Introduction

Surface micromachining is a process that uses thin film layers deposited on the surface of a substrate to construct structural components for MEMS. Unlike bulk micromachining that builds components within a substrate, surface micromachining builds on top of the substrate. The scanning electron microscope (SEM) image shows microgears that were fabricated using surface micromachining.

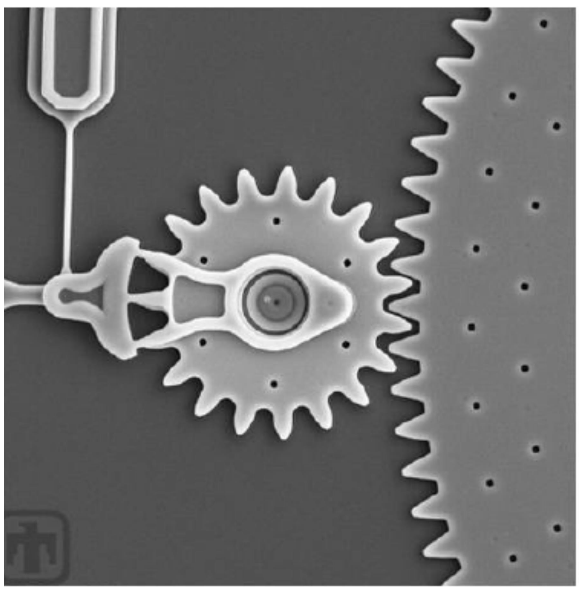


Figure 4.1: microgears that were fabricated using surface micromachining

These gears are very thin, 2 to 3 microns in thickness (or height), but can be hundreds of microns wide. Each gear tooth is smaller than the diameter of a red blood cell (8 to 10 microns). These gears rotate above the surface of the substrate. [SEM courtesy of Sandia National Laboratories]

Surface micromachining uses many of the same techniques, processes, and tools as those used to build integrated circuits (ICs) or more specifically CMOS (Complementary Metal Oxide Semiconductor) components. This process is used to fabricate micro-size components and structures by depositing, patterning, and etching a series of thin film layers on a silicon substrate.

This creates an ideal situation for integrating microelectronics with micromechanics. Electronic logic circuits can be fabricated at the same time and on the same chip as the mechanical devices. The 3-axes MEMS accelerometer in figure 4.2, shows three surface micro-machined accelerometers (mechanical components) on the same chip as their electronic control circuits.

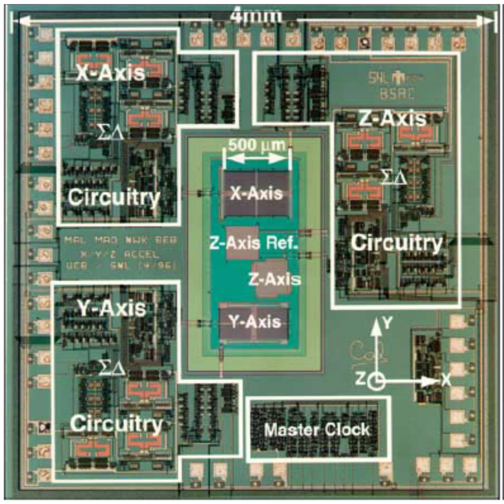


Figure 4.2: *Integrated 3-axes silicon microaccelerometer [Image courtesy of Sandia National Laboratories]*

4.3 Difference between CMOS fabrication and surface micromachining

The main difference between CMOS fabrication and surface micromachining is that the circuits constructed for CMOS allow for the movement or flow of electrons while the structures constructed with surface micromachining (e.g., cantilevers, gears, mirrors, switches) move matter. In order to move matter and to create moveable structures, spaces must be incorporated between moveable components during the fabrication process. For example, optical flip mirrors *(left of figure 4.3)* cannot move nor can a gear *(right of figure 4.3)* rotate on an axis unless there is space to allow for movement.

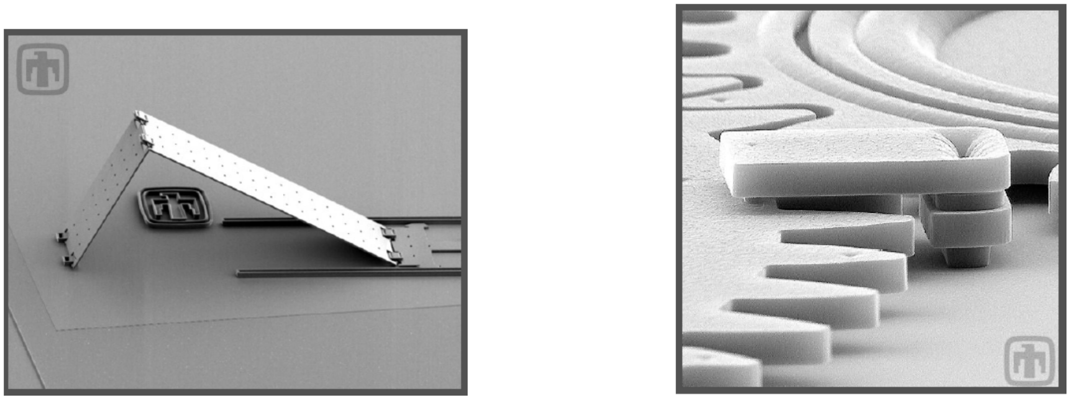


Figure 4.3: *Pop-up Mirror (left) and geartrain with alignment pin (right) [SEM images courtesy of Sandia National Laboratories]*

The spaces between components are fabricated using sacrificial layers. A sacrificial layer is deposited between two structural layers to provide the needed gap. Once the device is complete and all of the structural layers are formed, the sacrificial layers are removed, *releasing* the component(s) so that it is free to move. The graphic in figure 4.4, shows the construction of a micro-cantilever. A sacrificial layer is deposited on top of the substrate. A structural layer is then deposited on top of the sacrificial layer. Once the structure is defined and etched, the sacrificial layer is removed; the cantilever is *released* and is free to flex.

Some of the moving parts in the structural layers are so thin (2 to 3 microns) and have such a low aspect ratio (ratio of height to width) that they are sometimes referred to as “2.5 D” rather than 3 D.

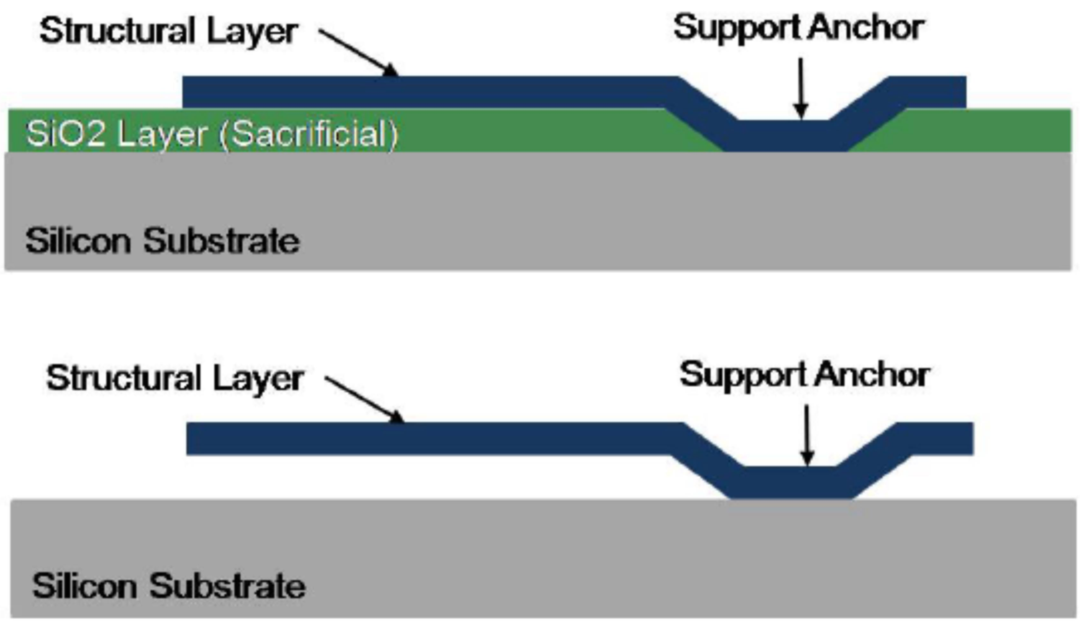


Figure 4.4: Surface micromachining of a micro-cantilever

4.5 Substrates for Surface Micromachining

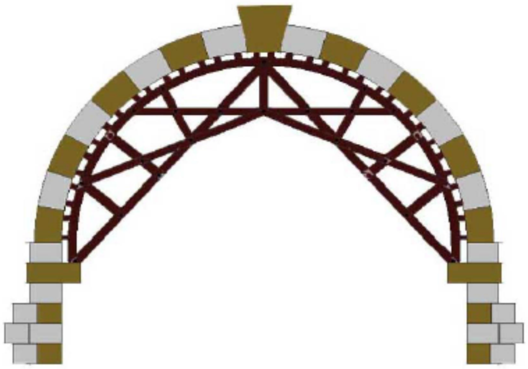
Surface micromachining is based on the deposition and etching of alternating structural and sacrificial layers on top of a substrate. The most commonly used substrate is silicon; however, less expensive substrates such as glass and plastic are also used. Glass substrates are used for MEMS applications such as DNA microarrays, implantable sensors, components for flat screen displays, and solar cells. Plastic substrates are used for various microfluidics applications and bioMEMS applications as well as for the fabrication of surface micromachined beams, diaphragms and cantilevers.

4.6 What is a Sacrificial Layer?

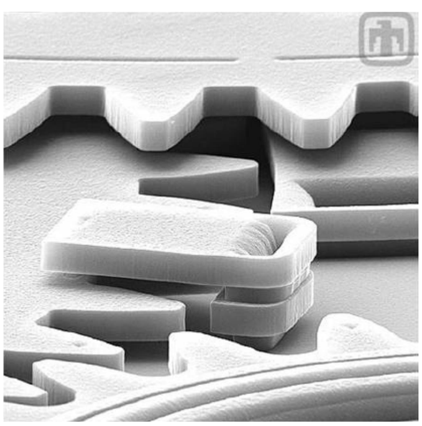
Complicated components such as movable gear transmissions and chain drives can be constructed using surface micromachining because of its use of the sacrificial layer. Let's take a look at how sacrificial devices are used to construct macro-size structures.

The image to the right shows a cross-sectional view of a keystone bridge. This structure is made by first constructing a wooden scaffold. Cut stones are placed on top of the scaffold, following its outline. The final stone at the apex is called the keystone, thus the name – Keystone bridge. Once this stone is in place, the scaffolding is removed and the bridge remains in place.

The scaffold is only used to provide support and shape during the construction process, and then it is sacrificed (removed). Thus the term sacrificial layer.



When constructing MEMS there are many possible combinations of sacrificial and structural layers. The combination used is dependent upon the device(s) being constructed. Below are two surface micromachined MEMS that require different process flows. Notice the layering required for the gears with its alignment clip figure 4.5a *(left)* vs. the pop-up mirror (right) figure 4.5b. Obviously, the gears would require more structural/sacrificial layers than the pop-up mirror.



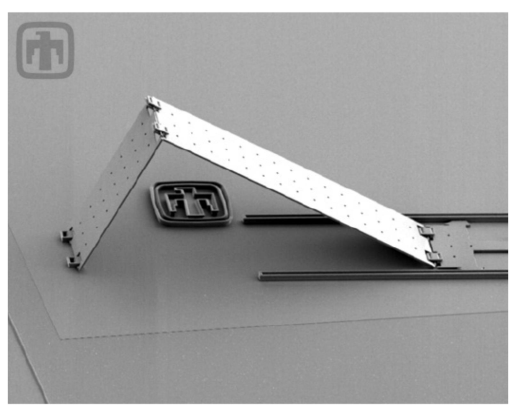


Figure 4.5a: gears **Figure 4.5b**: pop-up mirror

[SEM images courtesy of Sandia National Laboratories]

4.7 Surface Micromachining Materials

Materials used in surface micromachining are generally the same as those used in CMOS processing techniques but they serve a different function in the mechanical components.

- Silicon dioxide (SiO_2 or oxide) is the film most commonly used as a sacrificial layer and hard mask.
- Polysilicon crystalline (poly) is the most commonly used film as a structural layer.
- Silicon nitride is a thin film used for membranes (in devices such pressure sensors), as insulating material, and as a hard mask.
- Self-assembled monolayer (SAM) coatings are deposited at different steps to make the surfaces hydrophobic, and to reduce friction and wear of rubbing parts.

4.8 Surface Micromachining Layers and Processes

Silicon Dioxide (SiO_2 or oxide)

The first step of surface micromachining is to grow a thin film of silicon dioxide into the surface of the silicon wafer (substrate). This first SiO_2 layer acts as an insulator and a scaffold (space). It is thermally grown in a thermal oxidation furnace. The picture on the left is of a six (6) process chamber horizontal oxidation furnace. The graphic on the right illustrates the components of each chamber. This is a batch process; therefore, several cassettes (boats) of wafers are processed at one time.

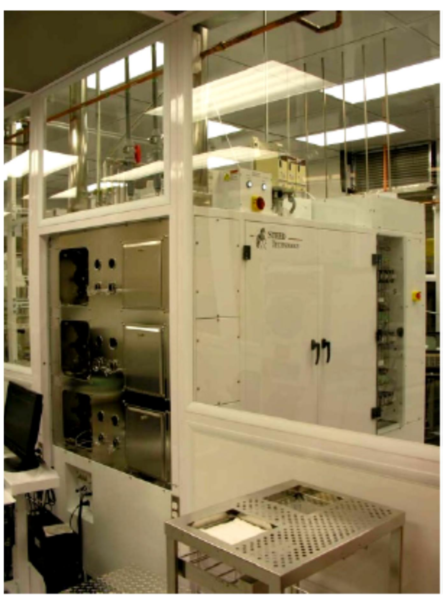


Figure 4.6: *Horizontal Thermal Oxidation Furnace [Photo courtesy of University of New Mexico, Manufacturing Training and Technology Center (UNM/MTTC)]*

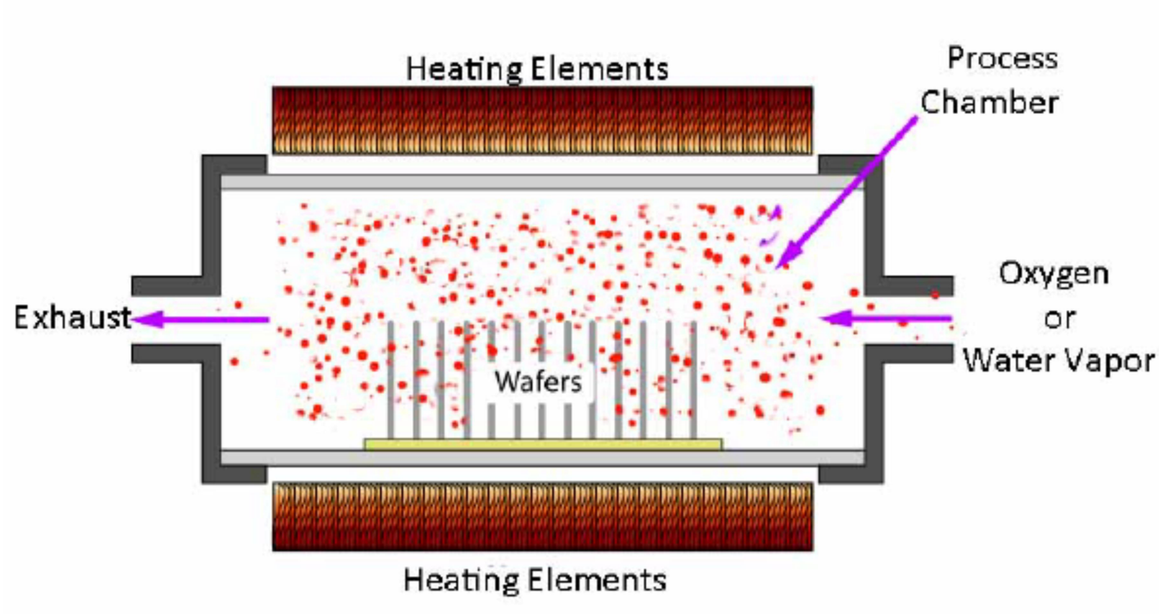


Figure 4.7: *Diagram of an oxidation furnace's process chamber*

Two oxidation methods are used in thermal oxidation: dry and wet oxidation. Dry oxidation uses oxygen gas (O2) to form SiO2.

Si (solid) + O_2 (gas) $\longrightarrow$ SiO_2 (solid)

Wet oxidation uses steam or water vapour to form SiO_2.

Si (solid) + $2H_2O$ (vapour) $\longrightarrow$ SiO_2 (solid) + $2H_2$ (gas)

In both processes, dry and wet, the process temperature affects the rate of oxidation (the rate at which the SiO_2 layer grows). The higher the temperature, the greater the oxidation rate (amount of oxide growth / time). Also, wet oxidation has a higher oxidation rate than dry oxidation at any given temperature. This effect can be seen in the oxidation of iron and the formation of rust (iron oxide). Rust grows much faster in humid climates than in dry climates.

4.9 Building a MEMS using Surface Micromachining

The linkage system in the graphic is an example of a surface micromachining process that requires several structural and sacrificial layers. During fabrication, the sacrificial oxide layers define the components' topographical shapes (structural layers) and the vertical spaces between them. Let's take a look at how such a device is fabricated.

- ♣ A sacrificial (oxide) layer is first deposited on the substrate.

- ♣ The first polysilicon layer is deposited on top of the oxide layer.
- ♣ This polysilicon layer is patterned and etched and forms the first set of cantilevers.
- ♣ A second oxide layer is deposited on top of the etched polysilicon layer.
- ♣ The second structural layer or polysilicon layer is deposited, patterned and etched. This forms the second set of cantilevers.
- ♣ The third oxide layer (sacrificial layer) is deposited.
- ♣ At this point, the surface has become extremely bumpy; therefore, this oxide deposition is followed by a CMP.
- ♣ Two holes are etched through the top oxide layer providing an opening to the polysilicon layer below it. These holes are needed to begin forming the posts which support the cantilevers and allow rotation.
- ♣ To continue fabricating the posts, holes are etched into the second polysilicon layer.
- ♣ Another layer of oxide is deposited on the surface and into the two holes. This is the last sacrificial layer.
- ♣ The third polysilicon layer is now deposited, patterned and etched. This layer forms the top set of cantilevers.
- ♣ The oxide films below each structural layer provide the necessary space for the middle cantilever to move after all of the oxide layers have been removed and the cantilevers released.
- ♣ The last step of this process is to remove the oxide layers from between the structural layers using a wet etch process of a hydrofluoric acid (HF) solution.
- ♣ Once the sacrificial layers are removed, the middle cantilever is free to rotate.

These steps - 1) oxide deposition, 2) structural layer deposition, pattern, etch, 3) sacrificial etch – can be repeated several times when fabricating a complex moving structure such as the linkage system, a gear transmission, an accelerometer, and other MEMS devices.

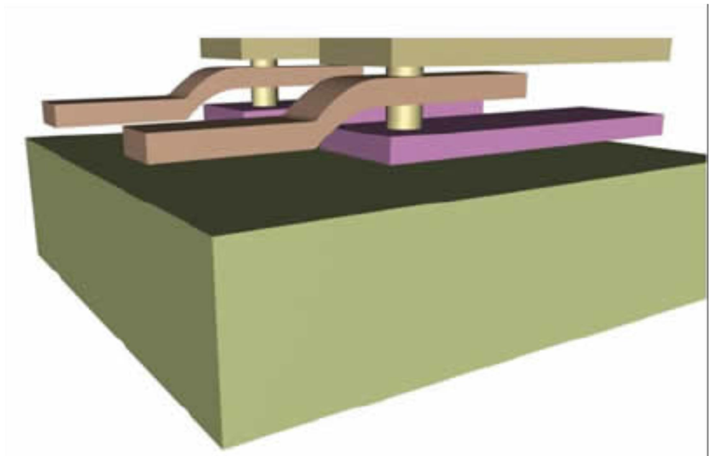


Figure 4.8: MEMS using Surface Micromachining

Note: An important advantage of using surface micromachining over bulk micromachining is that it is very compatible with CMOS processing. This compatibility allows mechanical devices to be built at the same time as the electronic logic circuits. Also, the cost of fabrication is generally lower since this technology uses the same equipment as the semiconductor industry. Many MEMS startup companies purchase used equipment from the semiconductor industry allowing for lower startup costs.

4.10: Disadvantage of Micromachining

- ✓ The downside of surface micromachining is that the mechanical components are very close together (a few microns) and they are flat. This can cause stiction. Stiction occurs when two, very flat surfaces come into contact and stick together; often, the two parts cannot be separated.
- ✓ A limitation of surface micromachining is that its processes can generally create only low aspect ratio devices, which is ideal for comb drives and gear drives. However, MEMS devices such as micro-channels and reservoirs require high aspect ratios; therefore, other micromachining processes are required.

4.11 Surface Micromachining Components

In spite of its limitations, surface micromachining is used to fabricate many MEMS components:

• Comb Drives *(Below left)*

• RF Switch *(center)*

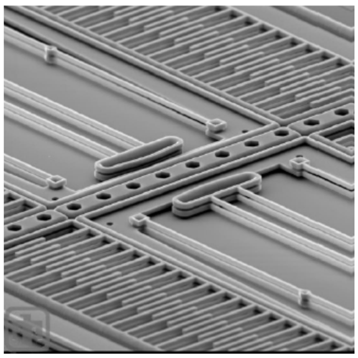
• Gears and chains *(Right)*

• Surface acoustical wave (SAW) sensors

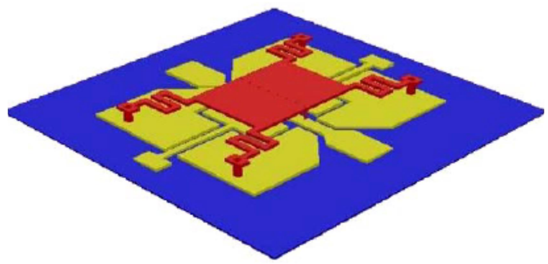
• Inertial sensors

• Cantilevers

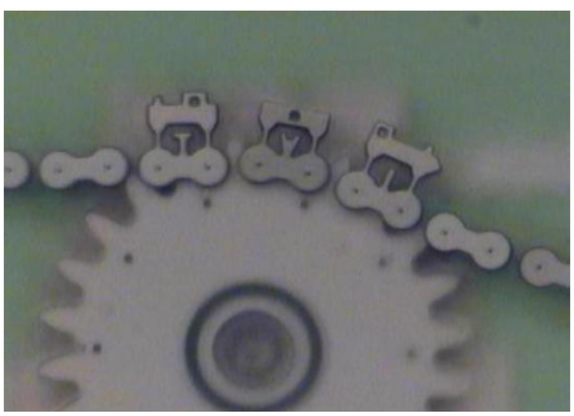
• Torsional Ratching Actuators (TRA)



Comb Drives



RF Switch



Chains and Gears

Figure 4.10: Components Fabricated Using Surface Micromachining